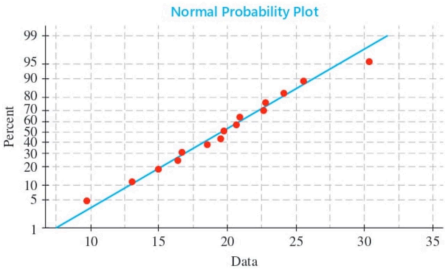
13.



Points on the normal probability plot approximately lies on a straight line

→ approximately normally distribute.

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20. Cerebral Blood Flow Cerebral blood flow (CBF) in the brains of healthy people is normally distributed with a mean of 74. A random sample of 25 stroke patients resulted in an average CBF of $\bar{x} = 69.7$ with a standard deviation of $s = 16$. Test the hypothesis that the average CBF for stroke patients is different from that of healthy people using $\alpha = .05$.

$H_0: \mu = 74 \quad H_a: \mu \neq 74$

$df = 25 - 1 = 24 \quad \bar{x} = 69.7$

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{69.7 - 74}{\frac{16}{\sqrt{25}}} = -1.343$$

$t_{24, 0.025} = 2.064$

$|t| = 1.343 < 2.064 \Rightarrow$ Non-reject H_0

	Pretreatment	Posttreatment	Follow-up
Sample Mean $\bar{x}$	20.3	26.6	27.7
Standard Deviation s	5.0	7.4	8.2
Sample Size n	21	21	20

a. Use a test of hypothesis to determine whether there is sufficient evidence to conclude that the true pretreatment mean is less than 25.

b. Construct a 95% confidence interval for the true posttreatment mean.

c. In Section 10.3, we will introduce small-sample techniques for making inferences about the difference between two population means. Without a formal statistical test, what might you conclude about the differences among the three sampled population means represented by the results in the table?

a. $H_0: \mu = 25 \quad H_a: \mu < 25$

$\bar{x} = 20.3 \quad s = 5 \quad n = 21 \quad df = 20$

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{20.3 - 25}{\frac{5}{\sqrt{21}}} = -4.3$$

查 $t_{20, 0.05} = 1.725$

$|t| = 4.3 > 1.725 \Rightarrow$ reject H_0

b. $\bar{x} \pm t \cdot \frac{s}{\sqrt{n}} = 26.6 \pm 2.086 \cdot \frac{7.4}{\sqrt{21}}$

$t_{20, 0.025} = 2.086$

$= (23.23, 29.97)$

c. since posttreatment and follow-up are higher than pretreatment ⇒ treatment is effective.

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35. Sodium Chloride Measurements of water intake, obtained from a sample of 17 rats that had been injected with a sodium chloride solution, produced a mean and standard deviation of 31.0 and 6.2 cubic centimeters (cm^3), respectively. Suppose that the average water intake for noninjected rats observed over a comparable period of time is 22.0 cm^3.

a. Do the data indicate that injected rats drink more water than noninjected rats? Test at the 5% level of significance.

b. Find a 90% confidence interval for the mean water intake for injected rats.

a. $H_0: \mu = 22 \quad H_a: \mu > 22$

$$t = \frac{31-22}{\frac{6.2}{\sqrt{17}}} \approx 5.99$$

查 $t_{16,\,0.05} = 1.746$

$5.99 > 1.746 \rightarrow$ reject H_0

b. $$\bar{x} \pm t \cdot \frac{s}{\sqrt{n}} = 31 \pm 1.746 \cdot \frac{6.2}{\sqrt{17}}$$

$$= 31 \pm 2.625$$

$$= (28.375,\ 33.625)$$

Two-Sample Tests *Use the information provided in Exercises 7–8. State the null and alternative hypotheses for testing for a significant difference in means, calculate the pooled estimate of σ^2, the associated degrees of freedom, and the observed value of the t statistic. What is the rejection region using $\alpha = .05$? What is the p-value for the test? What can you conclude from these data?*

7.

Population 1	12	3	8	5
Population 2	14	7	7	9	6

8.

Population	1	2
Sample Size	16	13
Sample Mean	34.6	32.2
Sample Variance	4.8	5.9

7. $n_1 = 4,\ \bar{x}_1 = 7,\ s_1^2 = 15.33$

$n_2 = 5,\ \bar{x}_2 = 8.6,\ s_2^2 = 10.3$

$$S_p^2 = \frac{(4-1)\cdot 15.33 + (5-1)\cdot 10.3}{7} = 12.456$$

$$T = \frac{7 - 8.6}{\sqrt{12.456 \cdot \left(\frac{1}{4} + \frac{1}{5}\right)}} = -0.676$$

$t_{7,\,0.025} =$

$|T| = 0.676 < 1.895 \rightarrow$ Non-reject H_0

8. $n_1 = 16,\ \bar{x}_1 = 34.6,\ s_1^2 = 4.8$

$n_2 = 13,\ \bar{x}_2 = 32.2,\ s_2^2 = 5.9$

$$S_p^2 = \frac{15 \cdot 4.8 + 12 \cdot 5.9}{27} = 5.289$$

$$T = \frac{34.6 - 32.2}{\sqrt{5.289\left(\frac{1}{16} + \frac{1}{13}\right)}} = 2.795$$

$t_{27,\,0.025} = 2.052$

$T > 2.052$

$\rightarrow$ reject H_0

advancement test recorded. The summarized data follow.

	① iPad	② Textbook
Mean	86.4	79.7
Standard Deviation	8.95	10.7
Sample Size	10	10

a. Use the summary data to test for a significant difference in advancement scores for the two groups using $\alpha = .05$.

b. Find a 95% confidence interval for the difference in mean scores for the two groups.

c. In light of parts a and b, what can we say about using iPads versus traditional textbooks in teaching algebra at the middle school level?

a. $n_1 = 10$、$\bar{x}_1 = 86.4$、$s_1^2 = 8.95^2$

$n_2 = 10$、$\bar{x}_2 = 79.7$、$s_2^2 = 10.7^2$

$H_0: \mu_1 - \mu_2 = 0 \quad H_a: \mu_1 - \mu_2 \neq 0$

$$t = \frac{86.4 - 79.7 - 0}{\sqrt{\frac{8.95^2}{10} + \frac{10.7^2}{10}}} = 1.518$$

$t_{18, 0.025} = 2.101$ → Non-reject H_0

b. $(\bar{x}_1 - \bar{x}_2) \pm t_{18, 0.025} \cdot 4.113$

$= (-2.568, 15.968)$

c. 95% CI 中包含"0"，無法證明推論顯著